MACRO L3 - 2014-2015

Lecture 5

F. PORTIER

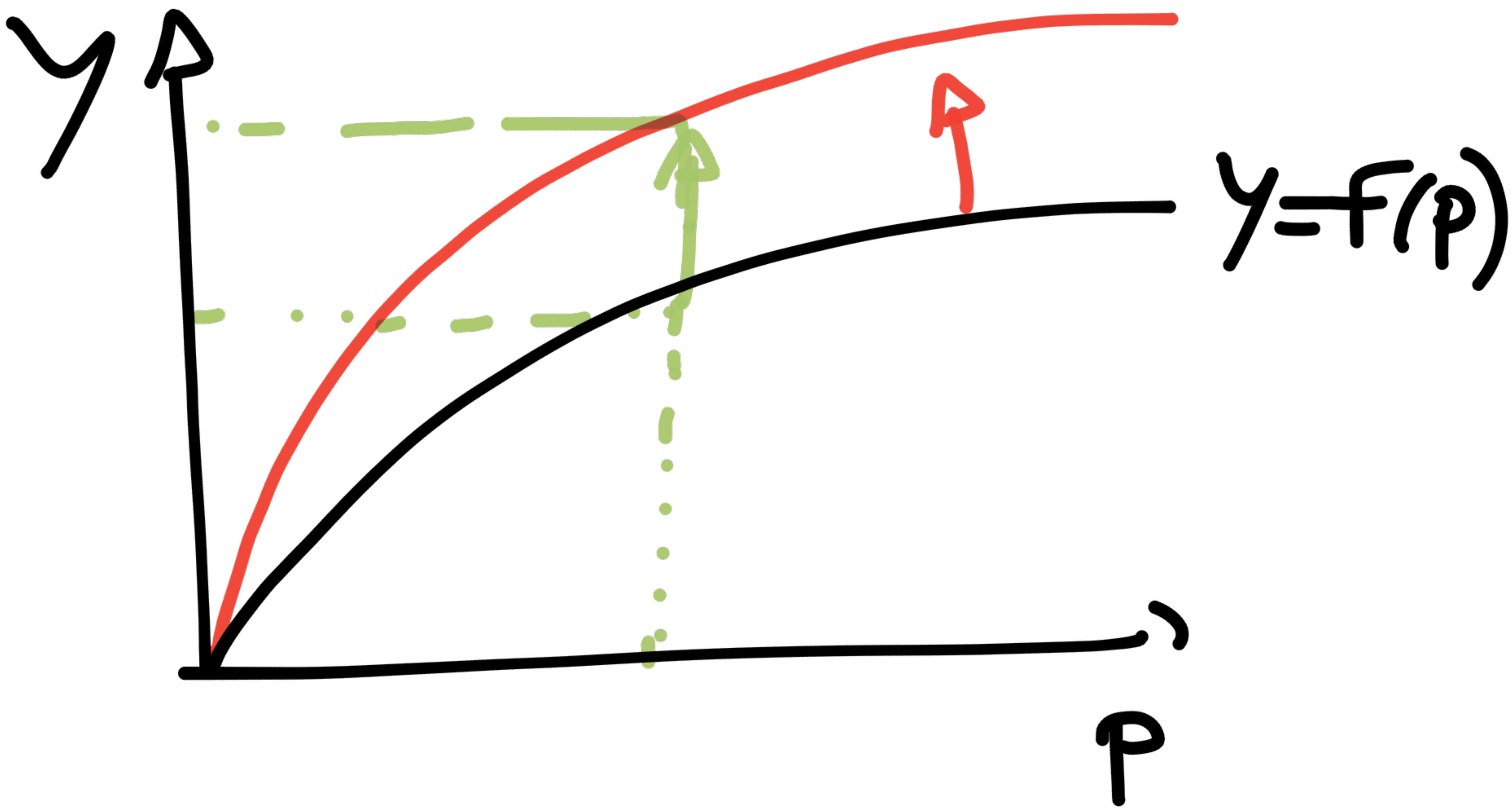
Session 3

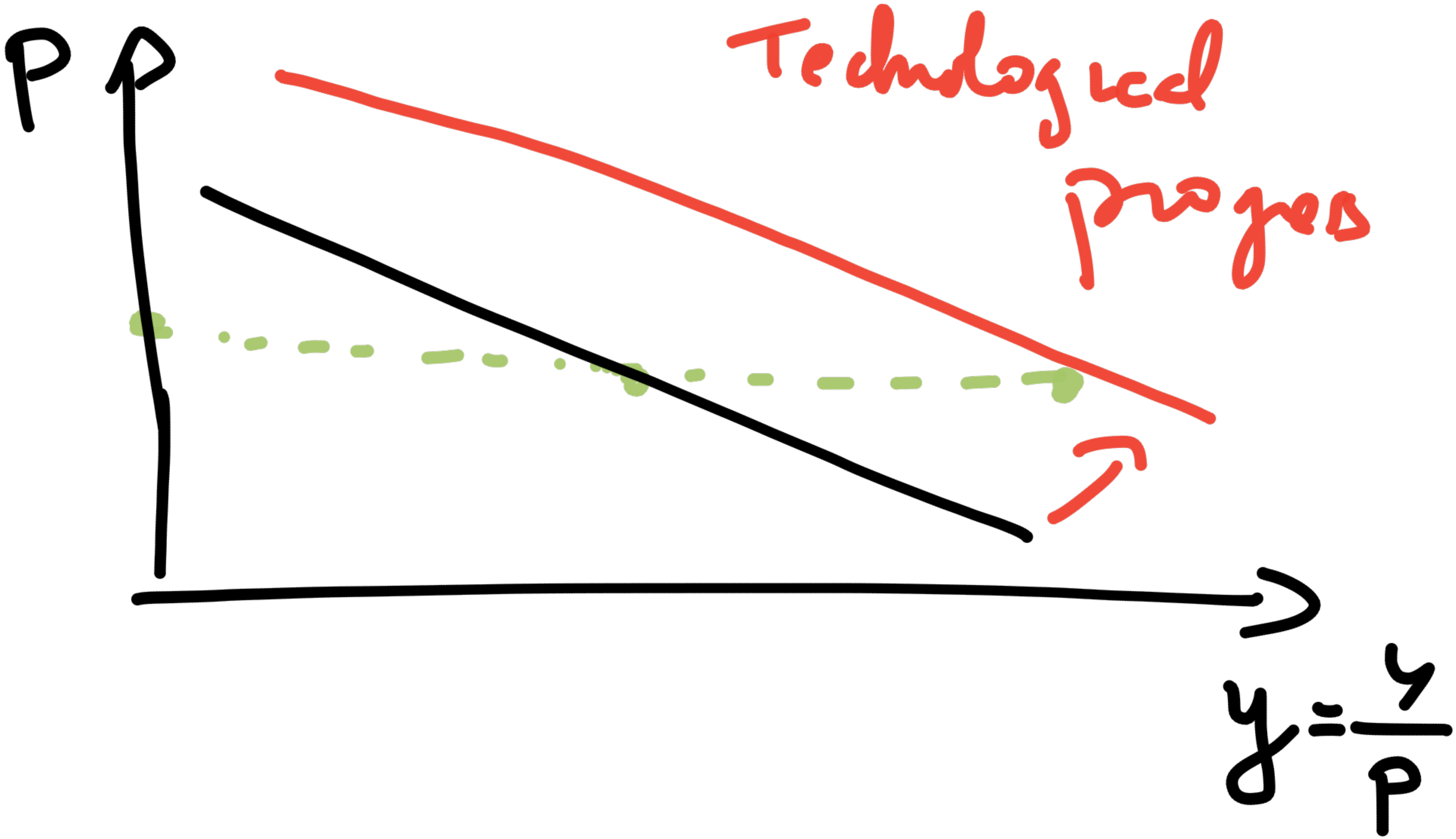
These notes are those I wrote during class. They are posted "for the record".

Session 3

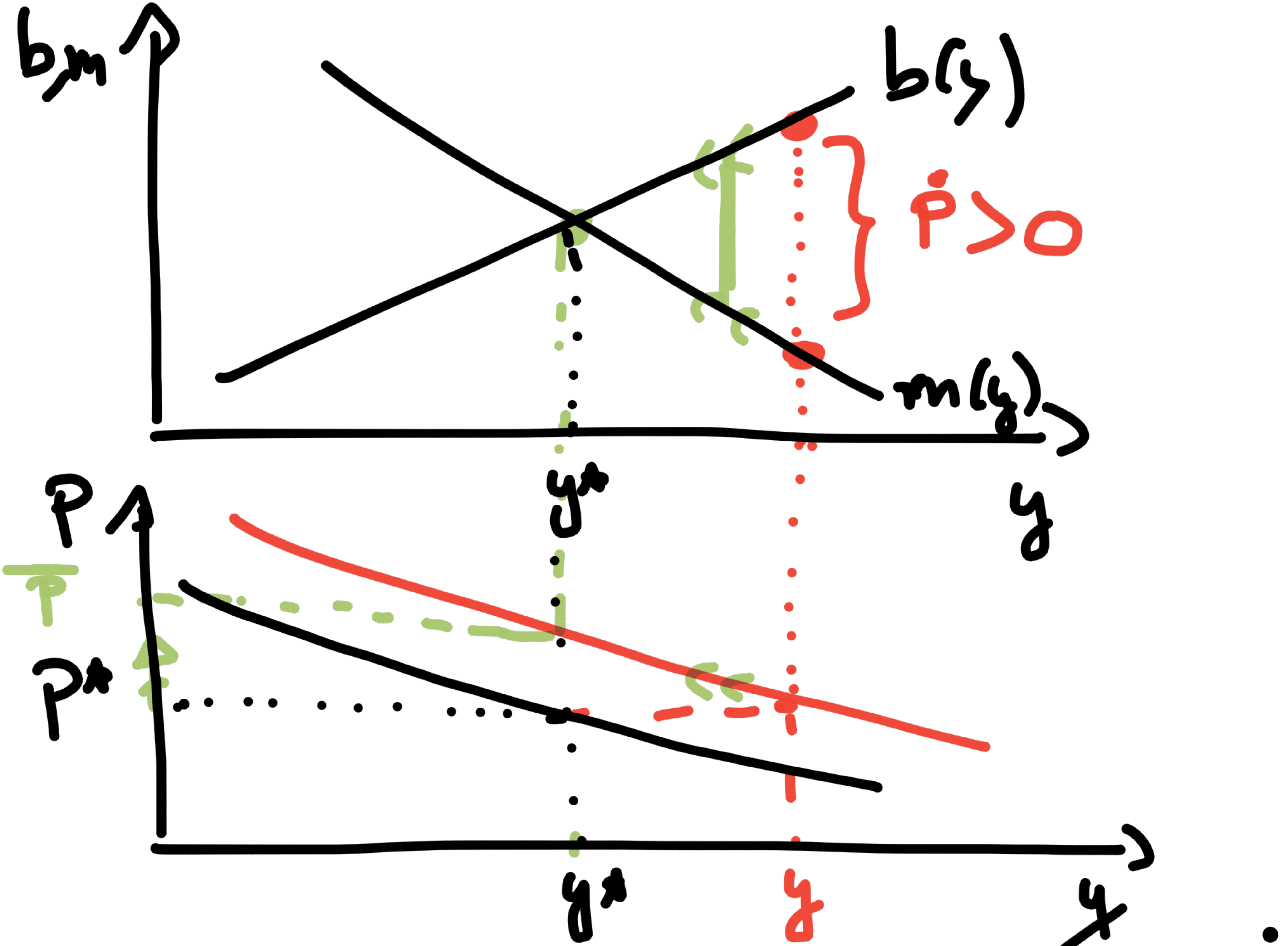
Nov 10, 2014

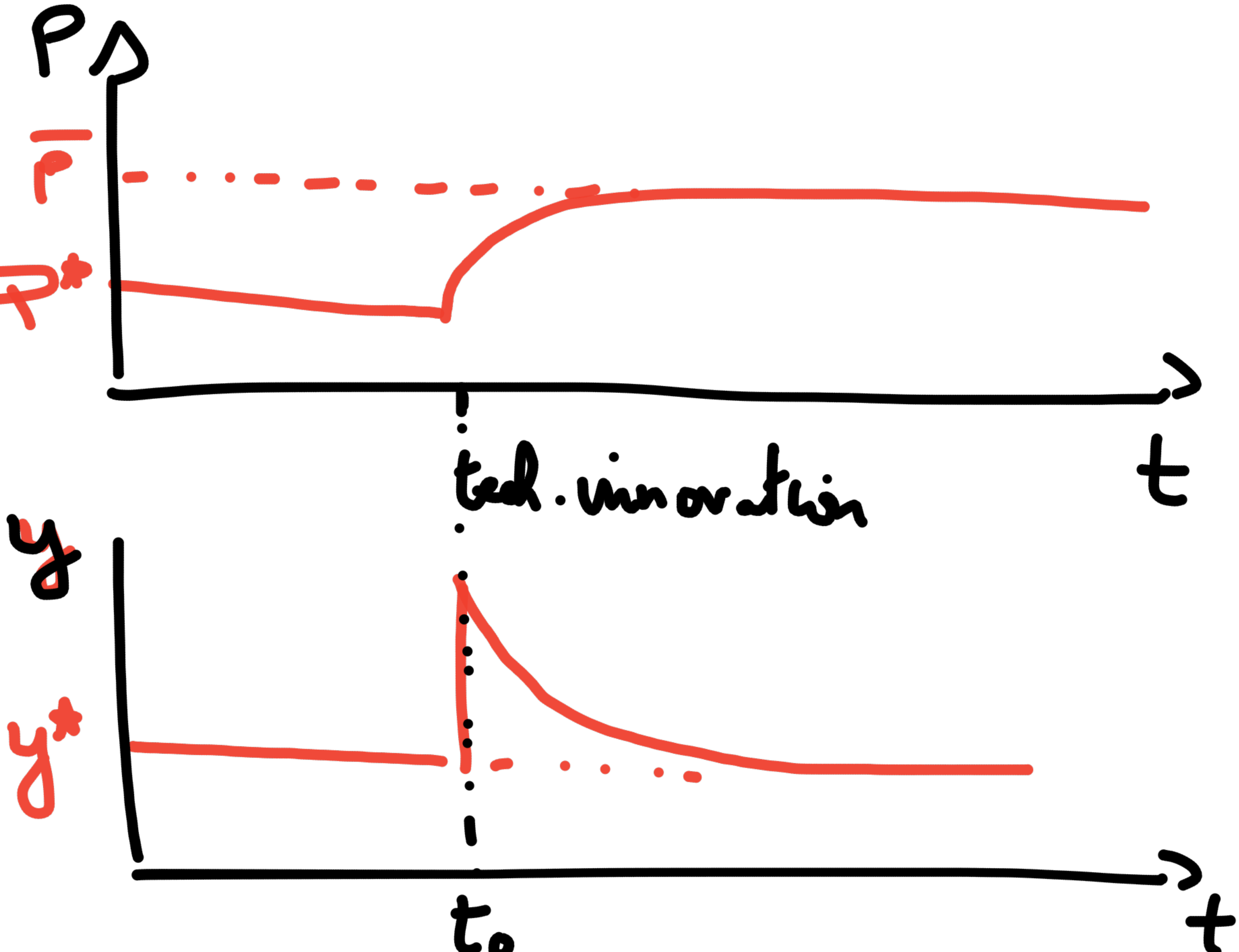
Technological progress





P
Technological
progress
y = Y/P


b,m
b(y)
Ṗ>0
m(y)
y
y*
P
P̄
P*
y*
y
y



P
$\bar{P}$
P^*
t
tech. innovation
y
y^*
t_0
t

$L =$ Lifetime of someone born in date 0

$$P(L > \ell) = e^{-m\ell} = \frac{1}{e^{m\ell}}$$

↳ the probability that lifetime L is greater than ℓ

[check $\ell = 0$, $\ell \to +\infty$]

Cumulative density function of L

$F(L) = P(L \leq \ell)$ (by definition of the cdf)

$= 1 - P(L > \ell)$

$\Rightarrow \boxed{F(\ell) = 1 - e^{-m\ell}}$

probability density function

(p.d.f) $f(\ell)$

$$f(\ell) = F'(\ell) = m e^{-m\ell}$$

Discrete to continuous

Discrete:

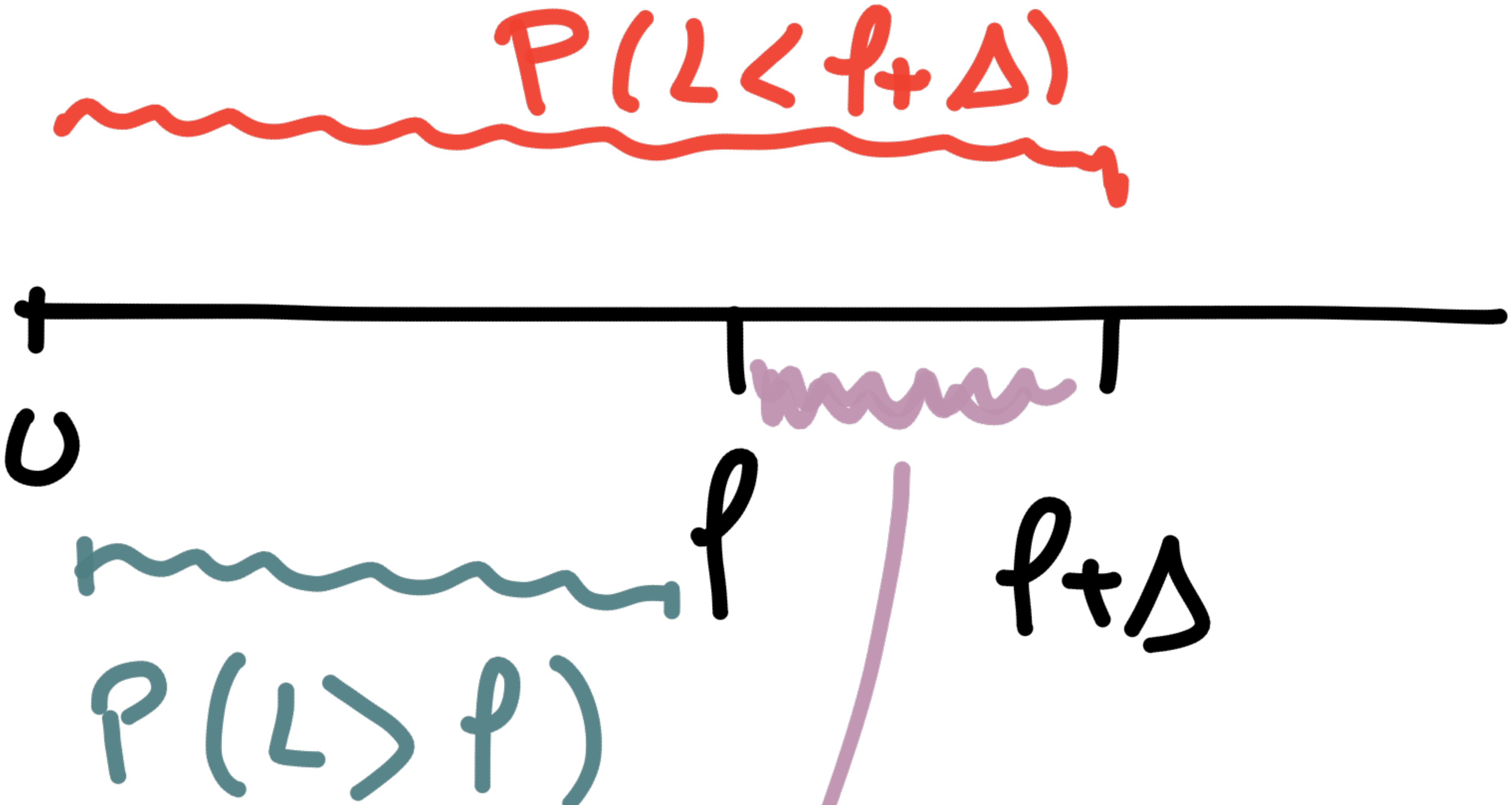
reduce Δ by 2

reduce Δ → 0

Δ = time interval (1 year, 1 day, 1 min)

From the point of view of date 0, probability of dying between age ℓ and age $\ell + \Delta$?

$$P(\ell < L < \ell + \Delta)\ ?$$



$$P(L < \ell+\Delta) - P(L > \ell) = P(\ell < L < \ell+\Delta)$$

Analytically:

$$P(\ell < L < \ell + \Delta)$$

$$= F(\ell + \Delta) - F(\ell)$$

$$= \left(1 - e^{-m(\ell+\Delta)}\right) - \left(1 - e^{-m\ell}\right)$$

$$= e^{-m\ell} - e^{-m(\ell+\Delta)}$$

Taylor exp

↳ I want to compute this

Taylor exp around x_0 of $f(x)$

$$f(x) \approx f(x_0) + (x - x_0) \times f'(x_0)$$

Here :

$$\left.\begin{aligned} x &= l + \Delta \\ x_0 &= l \end{aligned}\right] \longrightarrow \text{Taylor exp around } \Delta = 0$$

$$f(x) = e^{-mx} \quad , \quad f'(x) = -m\, e^{-mx}$$

$-$

Taylor exp:

$$\underline{e^{-m(\ell+\Delta)}} \approx e^{-m\ell} + \Delta \times \left(-e^{-m\ell}\right) \times m$$

$$= \underline{e^{-m\ell} - \Delta e^{-m\ell} \times} m$$

Therefore

$$P(\ell < L < \ell+\Delta) = e^{-m\ell} - e^{-m(\ell+\Delta)}$$

$$= +m\Delta e^{-m\ell}$$

$$P(l < L < l+\Delta) = m\Delta e^{-ml}$$

<u>Take</u> $\Delta = 1$ year

<u>Take</u> $l = 0$

$P(0 < L < 1)$ = probability of dying within the year

$$= m \times e^{-m\times 0} = m$$

- Model of perpetual youth
- Nevertheless, expected life time is finite

expected life length at birth:

$$\mathcal{E} = \int_0^{\infty} \ell f(\ell)\, d\ell = \int_0^{\infty} \ell m e^{-m\ell}\, d\ell$$

$$\mathcal{E} = \int_0^\infty \ell\, m\, e^{-m\ell}\, d\ell$$

↳ let's compute $\mathcal{E}$

<u>**by parts**</u>

$u(\ell) = m\ell \qquad u'(\ell) = m$

$v'(\ell) = e^{-m\ell} \qquad v(\ell) = -\frac{1}{m} e^{-m\ell}$

$$\mathcal{E} = \int_0^\infty u v'\, d\ell = \left[u v \right]_0^\infty - \int_0^\infty u' v\, d\ell$$

$$\mathcal{E} = \left[m\ell \times \left(\frac{-1}{m}\right) e^{-m\ell} \right]_0^\infty - \int_0^\infty m \left(-\frac{1}{m} e^{-m\ell}\right) d\ell$$

$$= \left[-\ell e^{-m\ell} \right]_0^\infty + \int_0^\infty e^{-m\ell} \, d\ell$$

$$= \left[-\ell e^{-m\ell} \right]_0^\infty + \left[-\frac{1}{m} e^{-m\ell} \right]_0^\infty$$

$$= 0 + 0 + 0 + \frac{1}{m} = \frac{1}{m}$$

life expectancy is finite.